

PrimeTrade.ai Hiring Assignment

Relationship Between Bitcoin Market Sentiment and Trader Performance

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Executive Summary

This project investigates the relationship between Bitcoin market sentiment and historical trader performance using two datasets: Hyperliquid historical trading data and the Bitcoin Fear & Greed Index.

The primary objective is to understand whether prevailing market sentiment influences trader profitability, trading behavior, and overall market participation. By integrating market sentiment with historical trading records, the analysis provides insights into how psychological market conditions affect trading outcomes and highlights opportunities for improving trading strategies.

The project involved data cleaning, preprocessing, dataset integration, exploratory data analysis (EDA), statistical analysis, and visualization to uncover meaningful relationships between market sentiment and trader performance.

Problem Statement

Cryptocurrency markets are highly influenced by investor psychology. Fear and greed often drive market movements, leading traders to behave differently under varying market conditions.

This project aims to answer the following questions:

- Does Bitcoin market sentiment affect trader profitability?
- Which market conditions produce better trading performance?
- Do traders behave differently during Fear and Greed periods?
- Which cryptocurrencies generate the highest cumulative profits?
- Can sentiment information be incorporated into better trading strategies?

Datasets Used

Dataset 1 – Historical Trader Data

The historical trader dataset contains transaction-level information including:

- Trader Account
- Cryptocurrency (Coin)
- Execution Price

- Trade Size (Tokens)
- Trade Size (USD)
- Buy/Sell Side
- Timestamp
- Closed Profit & Loss (Closed PnL)
- Trading Fee
- Position Information

This dataset forms the basis for evaluating trader performance.

Dataset 2 – Bitcoin Fear & Greed Index

The Fear & Greed Index represents the daily emotional state of the cryptocurrency market.

Each day is classified into categories such as:

- Extreme Fear
- Fear
- Neutral
- Greed
- Extreme Greed

The dataset also includes a numerical sentiment score used to represent market conditions.

Methodology

The project followed a structured analytical workflow consisting of the following stages.

1. Data Loading

Both datasets were imported into Python using the Pandas library.

2. Data Cleaning

The following preprocessing steps were performed:

- Removed duplicate records
- Checked for missing values
- Converted timestamp columns into datetime format
- Converted numerical columns into appropriate numeric data types
- Created a common Date column for integration

3. Data Integration

The two datasets were merged using the Date column, enabling each historical trade to be associated with the prevailing Bitcoin market sentiment on that day.

This integration allowed market psychology to be analyzed alongside trader performance.

4. Exploratory Data Analysis

The merged dataset was analyzed using descriptive statistics and visualizations to understand:

- Market sentiment distribution
- Profitability patterns
- Trading activity
- Coin performance
- Trader performance
- Win rates
- Trade size distribution
- Numerical correlations

Visual Analysis

The notebook includes multiple visualizations designed to answer different business questions.

Market Sentiment Distribution

This visualization shows the number of trades executed under different market sentiment conditions.

Purpose:

- Understand trader participation during Fear and Greed periods.
- Identify whether traders become more active during particular market conditions.

Average Closed Profit by Sentiment

This analysis compares average trader profitability across different market sentiments.

Purpose:

- Measure whether specific market conditions are associated with better average returns.
- Evaluate the influence of investor psychology on profitability.

Profit Distribution

A box plot illustrates the spread of profits and losses under each market sentiment category.

Purpose:

- Detect variability in trader performance.
- Identify outliers and extreme profit/loss events.

Buy vs Sell Analysis

The Buy vs Sell visualization compares trading direction across market sentiments.

Purpose:

- Determine whether traders exhibit directional bias under different emotional market conditions.
- Understand behavioral changes during Fear and Greed.

Coin-wise Profitability

The project ranks cryptocurrencies based on cumulative Closed PnL.

Purpose:

- Identify consistently profitable assets.
- Understand where trading activity generates the highest returns.

Top Performing Traders

Historical trader accounts are ranked based on cumulative profitability.

Purpose:

- Identify high-performing traders.
- Observe the distribution of trading success across participants.

Correlation Analysis

A correlation heatmap measures relationships between numerical trading variables.

Purpose:

- Detect variables associated with profitability.
- Explore relationships between trade size, execution price, fees, and Closed PnL.

Winning Trade Percentage

This analysis calculates the proportion of profitable trades under each market sentiment.

Purpose:

- Evaluate trading success rates across different market conditions.
- Compare consistency rather than total profitability.

Trade Size vs Closed PnL

A scatter plot with a trend line examines the relationship between trade size and realized profit.

Purpose:

- Determine whether larger trades consistently produce higher profits.
- Identify the presence of any linear relationship.

Key Findings

The analysis produced several important observations.

1. Market sentiment influences trading behaviour.

Trading activity changes across different sentiment categories, suggesting that investor psychology impacts market participation.

2. Profitability varies across sentiment categories.

Average Closed PnL differs between Fear, Neutral, and Greed conditions, indicating that market sentiment has a measurable effect on trading performance.

3. Cryptocurrency trading remains highly volatile.

Profit distributions show considerable variation, with both substantial gains and significant losses observed across all sentiment categories.

4. Larger trades do not necessarily produce higher profits.

The scatter plot demonstrates that increasing trade size alone does not guarantee improved trading outcomes.

Successful trading appears to depend on multiple factors beyond position size.

5. Certain cryptocurrencies consistently outperform others.

Coin-wise analysis shows that a relatively small number of assets contribute disproportionately to total profitability.

These assets may warrant additional research for future strategy development.

6. Trading performance is concentrated among a subset of traders.

The top-performing accounts contribute a significant share of cumulative profits.

This suggests differences in strategy quality, experience, or risk management.

7. Profitability depends on multiple interacting variables.

Correlation analysis indicates that no single numerical feature completely explains Closed PnL.

Trading performance results from a combination of market conditions, trade execution, position sizing, and trader decision-making.

Business Insights

The findings have practical implications for cryptocurrency trading.

- Market sentiment should be considered as an additional trading indicator rather than used in isolation.
- Monitoring Fear & Greed levels can improve awareness of prevailing market conditions.
- Position sizing should be adjusted according to market uncertainty.
- Historical trader performance can help identify effective trading behaviours.
- Diversification across consistently profitable assets may improve long-term performance.

Recommendations

Based on the analysis, the following recommendations are proposed.

1. Integrate Bitcoin market sentiment into trading decision frameworks.
2. Apply stricter risk management during Fear and Extreme Fear periods.
3. Avoid assuming that larger trade sizes will generate greater returns.
4. Focus on historically profitable cryptocurrencies while continuously monitoring market conditions.
5. Combine sentiment analysis with technical indicators and proper position sizing.
6. Regularly evaluate historical trading performance to refine trading strategies.

Limitations

The analysis has several limitations.

- The study is based on historical data and may not predict future market behaviour.
- Market sentiment is only one of many variables influencing profitability.
- External macroeconomic events were not included in the analysis.
- Trading strategy details for individual accounts were unavailable.

Future work could incorporate additional market indicators such as volatility, trading volume, funding rates, and technical indicators.

Conclusion

This project successfully integrated Bitcoin market sentiment with historical trading data to investigate how market psychology influences trader performance.

The analysis demonstrates that sentiment plays a meaningful role in trading behaviour, profitability, and market participation. However, successful trading cannot be attributed solely to market sentiment. Instead, profitability appears to result from the interaction of market conditions, disciplined risk management, asset selection, and effective trading strategies.

Overall, the findings suggest that incorporating market sentiment into a broader data-driven trading framework can support more informed decision-making and improved risk management in cryptocurrency markets.